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Daniel Arnold

APPOINTMENTS

Lawrence Livermore National Laboratory, Livermore, Ca. — *Lead Power Systems Engineer*

May 2025 - Present

- Conduct research at the intersection of electric power systems, machine learning, statistical modeling, optimization, and control theory.
- Principal Investigator, responsible for leading and managing complex research projects. Duties include fundraising, project planning, project execution, interacting with program sponsors and other stakeholders.
- Develop machine learning, optimization, and statistical models and software for electric power systems and critical infrastructure.
- DOE Q-level security clearance

Lawrence Berkeley National Laboratory, Berkeley, Ca. — *Research Scientist*

July 2017 - May 2025

- Deputy Cybersecurity R&D Lead, conducted research on the cybersecurity of electric power systems.
- Developed artificial intelligence and optimal control approaches for managing distributed generation sources to increase the cyber-resiliency of the electric power distribution grid.

Civil and Environmental Engineering Dept., UC Berkeley, Berkeley, Ca. — *Adjunct Professor*

July 2023 - PRESENT

Civil and Environmental Engineering Dept., UC Berkeley, Berkeley, Ca. — *Lecturer*

Jan. 2018 - July 2023

Lawrence Berkeley National Laboratory, Berkeley, Ca. — *ITRI-Rosenfeld Postdoctoral Fellow*

Jan. 2016 - July 2017

- Develop data analysis pipeline in python for clustering and regression analysis of electric power distribution grid synchrophasor data.
- Develop model-free optimization algorithms for real-time control of distributed energy resources.

TEACHING

E93 - Energy Engineering Seminar (youtube)

- Organized speaker series which brings together industry leaders, researchers, and innovators in the energy sector to share insights on sustainable energy solutions, technological advancements, and policy implications.
- Fall 2026 focus on data center impacts, modeling, and optimization.

CE 295 - Data Science for Energy (https://ecal.berkeley.edu/ce295.html)

- Introduces data science fundamentals and programming techniques to graduate and upper division students, with energy focus
- Course content includes: dynamic systems modeling, state estimation, convex and stochastic optimization, machine learning, and optimal control
- Laboratory assignments applied to battery state of charge modeling, oil and natural gas extraction, power system optimization, building energy management, and optimal control of electric vehicles.

CE 191 - Engineering Systems Analysis
(https://ecal.berkeley.edu/ce191.html)

- Introduces optimization fundamentals and programming techniques to graduate and upper division undergraduate students.
- Course content: linear programming, quadratic programming, mixed integer programming, nonlinear programming, and dynamic programming.

EDUCATION

U.C. Berkeley, Berkeley, Ca. — *Ph.D Mechanical Engineering*

Sept. 2009 - Dec. 2015
Advisors: Duncan Callaway and David Auslander
Dissertation: Extremum Seeking Control of Distributed Energy Resources

U.C. San Diego, San Diego, Ca. — *M.S. Mechanical Engineering*

Sept. 2005 - Dec. 2006
Advisor: Miroslav Krstic

U.C. San Diego, San Diego, Ca. — *B.S. Mechanical Engineering (cum laude)*

Sept. 2001 - Sept. 2005

RELEVANT PROJECTS

Power, Resilience, and Infrastructure Security Modeling for Data Centers (PRISM-DC), PI — *$2,400,000 award*

June 2026 - May 2027

- Sponsor: Risk Management and Tools (RMT) program, U.S. Department of Energy, CESER office, and Critical Minerals and Energy Innovation (CMEI) office
- Objective: Develop statistical models of data center GPU population dynamics. Develop approaches for data center load forecasting and state estimation. Creation of a co-simulation framework to simulate data center dynamic behavior and impacts on the electric transmission system.

Mitigation via Analytics for Grid-Inverter Cybersecurity (MAGIC) Extension, PI — *$6,000,000 award*

Oct. 2026 - Sept. 2029

- Sponsor: Risk Management and Tools (RMT) program, U.S. Department of Energy, CESER office
- Objective: Develop machine learning algorithms for cyber attack detection and mitigation. Develop a co-simulation framework to model distributed energy resource behavior and electric grid information technology (IT) communications.

Mitigation via Analytics for Grid-Inverter Cybersecurity (MAGIC), PI — *$5,000,000 award*

Jan. 2024 - Dec. 2027

- Sponsor: Risk Management and Tools (RMT) program, U.S. Department of Energy, CESER office
- Objective: Creation of artificial intelligence-based cyber attack detection and mitigation algorithms for distributed energy resources in the electric power grid. Develop zero-shot grid foundational model for task transferability across unseen power grid networks.

SELECT PUBLICATIONS

Conference

S. Miguel, T. Wu, Y. Tao, A. Scaglione and D. Arnold, "Joint Detection of Cyber-Physical Attacks and False Data Injection Attacks: A Unified GCN Framework," *57th North American Power Symposium (NAPS)*, Storrs, CT, USA, 2025

Wu, T., Scaglione, A., Arnold, D., and Chen, T., “Transferable Learning of GCN Sampling Graph Data Clusters from Different Power Systems.” Allerton Conference, 2024.

Wu, T., Scaglione, A., and Arnold, D. “Constrained Reinforcement Learning for Dynamic Optimal Power Flow Control”, IEEE Power and Energy Society General Meeting (PESGM), 2023.

Roberts, C., Callaway, D. S., and Arnold, D., “Adaptive Control of Converter-Interfaced Generation for Cyber-Physical Security,” 56th Hawaii International Conference on System Sciences (HICSS), Maui, HI, Jan. 2023.

Wu, T., Scaglione, A., and Arnold, D. “Graph Convolutional Neural Network for the Control of Smart Inverters in Power Grids”, 58th Annual Allerton Conference on Communication, Control, and Computing (Allerton) , Sept. 2022.

D. Arnold, S. Ngo, C. Roberts, Y. Chen, A. Scaglione, S. Peisert, “Adam-based Augmented Random Search for Control Policies for Distributed Energy Resource Cyber Attack Mitigation”, American Control Conference (ACC), Atlanta, GA., June 2022, pp. 4559-4566.

C. Roberts, S. Ngo, A. Milesi, A. Scaglione, S. Peisert, and D. Arnold, “Deep Reinforcement Learning for Mitigating Cyber-Physical DER Voltage Unbalance Attacks", American Control Conference (ACC), New Orleans LA, 2021, pp. 2861-2867.

C. Roberts, S. Ngo, A. Milesi, S. Peisert, S. Saha, A Scaglione., N. Johnson, A. Kocheturov, D. Fradkin, and D. Arnold “Deep Reinforcement Learning for DER Cyberattack Mitigation", IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Tempe, AZ, USA, 2020, pp. 1-7.

Journal

T. Wu, A. Scaglione, S. Miguel and D. Arnold, "Universal Graph Convolutional Network: Zero-Shot Transfer Across Power Grid Topology Reconfigurations," in IEEE Transactions on Smart Grid, (early access)

Wu. T, Ravi, N., Scaglione, S., Peisert, S., and Arnold, D., “Optimal Control of Stochastic and Differentially Private EV Charging: A Scalable Learning Approach,” in IEEE Internet of Things Journal, vol. 12, no. 24, pp. 55061-55075, 15 Dec.15, 2025, https://doi.org/10.1109/JIOT.2025.3622650

Wu, T., Scaglione, A., and Arnold, D., “Constrained Reinforcement Learning for Predictive Control in Real-Time Stochastic Dynamic Optimal Power Flow,” in IEEE Transactions on Power Systems, vol. 39, no. 3, pp. 5077-5090, May 2024.

Wu, T., Scaglione, A., and Arnold, D. “Complex-Value Spatio-temporal Graph Convolutional Neural Networks and its Applications to Electric Power Systems AI,” IEEE Transactions on Smart Grid, 2023.

Wu, T., Scaglione, A., and Arnold, D., “Constrained Reinforcement Learning for Predictive Control in Real-Time Stochastic Dynamic Optimal Power Flow,” in IEEE Transactions on Power Systems, 2023.

Roberts, C., Arnold, D., and Callaway, D., “An Online Adaptive Damping Controller for Converter-Interfaced Generation.”, IEEE Transactions on Power Systems, 2023.

Losada Carreno, I., Saha, S. S., Scaglione, A., Arnold, D., Ngo, S., and Roberts, C., “Log(v) 3LPF: A Linear Power Flow Formulation for Unbalanced Three-Phase Distribution Systems,” IEEE Transactions on Power Systems, Vol. 38(1), Mar. 2022, pp. 100 - 113.

D. Arnold, S.Saha, S. Ngo, C. Roberts, A. Scaglione, N. Johnson, S. Peisert, and D. Pinney, “Adaptive Control of Distributed Energy Resources for

Distribution Grid Voltage Stability", IEEE Transactions on Power Systems, *Vol. 38(1), Mar. 2022, pp. 129 - 141.*

S. Saha. D. Arnold, A. Scaglione, E. Schweitzer, C. Roberts, S. Peisert, and N. Johnson, "Lyapunov Stability of Smart Inverters Using Linearized DistFlow Approximation", IET Renewable Power Generation, vol. 15, no. 1, pp. 114-126, 2021.

R. Dobbe, W. Van Westering, S. X. Liu, D. Arnold, D. S. Callaway and C. Tomlin, "Linear Single- and Three-Phase Voltage Forecasting and Bayesian State Estimation with Limited Sensing," IEEE Transactions on Power Systems, vol. 35, no. 3, pp. 1674-1683, May 2020.

E. Schweitzer, S. S. Saha, A. Scaglione, N. G. Johnson and D. Arnold, "Lossy DistFlow Formulation for Single and Multiphase Radial Feeders," IEEE Transactions on Power Systems, vol. 35, no. 3, pp. 1758-1768, May 2020.

R. Dobbe, O. Sondermeijer, D. Fridovich-Keil, D. Arnold, D. Callaway, C. Tomlin, "Towards Distributed Energy Services: Decentralizing Optimal Power Flow with Machine Learning", IEEE Transactions on Smart Grid, vol. 11, no. 2, pp. 1296-1306, March 2020.

M.D. Sankur, R. Dobbe, A. von Meier, E. Stewart, and D. Arnold, "Model-Free Optimal Voltage Phasor Regulation in Unbalanced Distribution Systems", vol. 11, no. 1, pp. 884-894, Jan. 2020.

C. Roberts, A. Scaglione, M. Jamei, R. Gentz, S. Peisert, E. Stewart, C. McParland, A. McEachern, and D. Arnold, "Learning Behavior of Distribution System Discrete Control Devices for Cyber-Physical Security", IEEE Transactions on Smart Grid, vol. 11, no. 1, pp. 749-761, Jan. 2020.

D. Arnold, M. Negrete-Pincetic, M. Sankur, and D. Callaway, "Model-Free Optimal Coordination of Distributed Energy Resources for Provisioning Transmission-Level Services" IEEE Transactions on Power Syst., vol. 33, issue 1, pp. 817-828, 2018.

D. Arnold, M. Negrete-Pincetic, M. Sankur, D. Auslander, and D. Callaway, "Model-Free Optimal Control of VAR Resources in Distribution Systems: An Extremum Seeking Approach," IEEE Transactions on Power Systems, vol. 31, issue 5, pp. 3583-3593, 2016.

C. Zhang, D. Arnold, N. Ghods, A. Siranosian, and M. Krstic, "Source Seeking with Nonholonomic Unicycle Without Position Measurement and with Tuning of Forward Velocity," Systems and Control Letters, vol. 56, pp. 245-252, 2007.

PATENTS

Granted

Arnold, D., Saha, D., Roberts, C., Ngo, S., Peisert, S., and Scaglione, S., (2026) System and method for control of distributed energy resources for distribution grid voltage stability (U.S. Patent No. 12531421). U.S. Patent and Trademark Office.

Arnold, D., and Sankur, M., (2024) Extremum seeking control of distributed energy resources with decaying dither and equilibrium-based switching (U.S. Patent No. 11881709). U.S. Patent and Trademark Office, Assignee: U.S. Department of Energy.